17/01/25 MIS

USN | | | | | | | | | | [23BSMA13]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2025

CALCULUS, LAPLACE TRANSFORMS AND MATRICES

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.
ii. Answer all questions

			Blooms Level	CO	Marks
		PART-A	**5X2=10**		
1.	**a.**	If $f = e^{-2x}siny + e^{4y}cosx$ then find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.	L2	CO1	**02**
	b.	If $\vec{F} = (x+y+1)\,i + j - (x+y)\,k$ then $curl\vec{F}$ ___	L1	CO1	**02**
	c.	$L[sinh\ at]$=___	L1	CO3	**02**
	d.	$L^{-1}\left[\frac{s+1}{s^2+4s+4}\right]$=___	L1	CO4	**02**
	e.	Find rank of the matrix $A = \begin{bmatrix} 2 & 3 & 5 & 1 \\ 1 & 2 & 3 & 2 \\ 1 & 3 & 4 & 5 \end{bmatrix}$	L1	CO1	**02**
		PART-B	**6X5=30**		
2.	**a.**	If $u = e^{ax+by}f(ax-by)$ then Prove that $b\frac{\partial u}{\partial x} + a\frac{\partial u}{\partial y} = 2abu$.	L3	CO2	**05**
		OR			
	b.	If $u = \frac{yz}{x}, v = \frac{zx}{y}, w = \frac{xy}{z}$ then show that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$.	L3	CO2	**05**
3.	**a.**	A particle moves along the curve $x = 1 - t^3, y = 1 + t^2$ & $z = 2t - 5$ (i) Determine its velocity and acceleration. (ii) Find the components of velocity and acceleration at $t = 1$ in the direction $2i + j + 2k$.	L3	CO1	**05**
		OR			
	b.	If $\vec{F} = \nabla(xy^3z^2)$ then find $div\vec{F}$ and $curl\vec{F}$ at the point $(1,-1,1)$.	L3	CO6	**05**
4.	**a.**	Evaluate $L\left[(t^2 - 6t + 9)e^{-(t-3)}u(t-3)\right]$.	L3	CO3	**05**
		OR			
	b.	Find the value of $\int_0^\infty te^{-3t}cos2t\ dt$ using Laplace transforms.	L3	CO3	**05**
5.	**a.**	Evaluate $L^{-1}\left[log\left(\frac{s+a}{s+b}\right)\right]$.	L3	CO4	**05**

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		OR			
	b.	Find the inverse Laplace Transform of $\frac{3}{s^2}+\frac{2e^{-s}}{s^3}-\frac{3e^{-s}}{s}$.	L3	CO4	**05**
6.	**a.**	Solve the system of given equations by Gauss-elimination method $x+y+z=9$, $x-2y+3z=8$ and $2x+y-z=3$.	L3	CO1	**05**
		OR			
	b.	Identify the rank of the following matrix by reducing into the row echelon form $$A=\begin{bmatrix}91 & 92 & 93 & 94 & 95\\ 92 & 93 & 94 & 95 & 96\\ 93 & 94 & 95 & 96 & 97\\ 94 & 95 & 96 & 97 & 98\\ 95 & 96 & 97 & 98 & 99\end{bmatrix}$$	L3	CO1	**05**
7.	**a.**	Write MATLAB code to determine the Inverse Laplace transform for the functions (i) $\frac{1}{(s-2)^2+16}$ (ii) $\frac{2s-5}{4s^2-25}$ (iii) $\frac{2s^2+3s-1}{s^2+3s+2}$.	L1	CO1	**05**
		OR			
	b.	Write MATLAB code to find the first and second-order partial derivatives of the function $f(x,y)=x^2+5xy-2y^2$.	L1	CO2	**05**
		PART-C	**6X10=60**		
8.	**a.**	Prove that $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=\frac{3}{(x+y+z)}$, where $u=log(x^3+y^3+z^3-3xyz)$ and hence show that $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u=\frac{-9}{(x+y+z)^2}$.	L3	CO2	**10**
		OR			
	b.	Investigate the extreme values of the given function $$x^3+3xy^2-3x^2-3y^2+4.$$	L3	CO6	**10**
9.	**a.**	Show that $\vec{F}$ is conservative force field $$\vec{F}=(2xy^2+yz)\,i+(2x^2y+xz+2yz^2)\,j+(2y^2z+xy)\,k$$ Also find its scalar potential.	L3	CO6	**10**
		OR			
	b.	Show that $\vec{F}=\frac{x\,i+y\,j}{x^2+y^2}$ is both solenoidal and irrotational.	L3	CO6	**10**
10.	**a.**	Given $f(t)=\begin{cases}E, & 0<t<a/2\\ -E, & a/2<t<a\end{cases}$ where $f(t+a)=f(t)$ then show that $L[f(t)]=\frac{E}{s}tanh(as/4)$.	L3	CO3	**10**

		OR			
	b.	Show that $\int_0^\infty t^3 e^{-t} sint = 0.$	L3	CO3	**10**
11.	**a.**	i. State convolution theorem. ii. Using convolution theorem find the inverse Laplace transform of $\frac{1}{(s^2+a^2)^2}$	L3	CO4	**10**
		OR			
	b.	Solve $y''' + 2y'' - y' - 2y = 0$, given $y(0) = y'(0) = 0$ and $y''(0) = 6$ by using Laplce transform.	L3	CO4	**10**
12.	**a.**	Investigate the Eigen values and the corresponding Eigen vectors of the matrix $A = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$	L3	CO5	**10**
		OR			
	b.	Investigate the values of λ and μ so that the system of equations $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$ and $2x + 3y + \lambda z = \mu$ will have **i.** no solution, **ii.** a unique solution **iii.** infinite number of solutions	L3	CO1	**10**
13.	**a.**	Write MATLAB code to find maxima and minima for function of two variables.	L1	CO6	**10**
		OR			
	b.	Write MATLAB program to perform the following operations on the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$, **i.** Compute the eigenvalues **ii.** Compute the eigenvectors **iii.** Obtain the diagonal matrix **iv.** Calculate A^4 using diagonalization	L1	CO5	**10**